Worksheet 4.5

Ionic bonding

Answers

Ions forming stable electron configurations

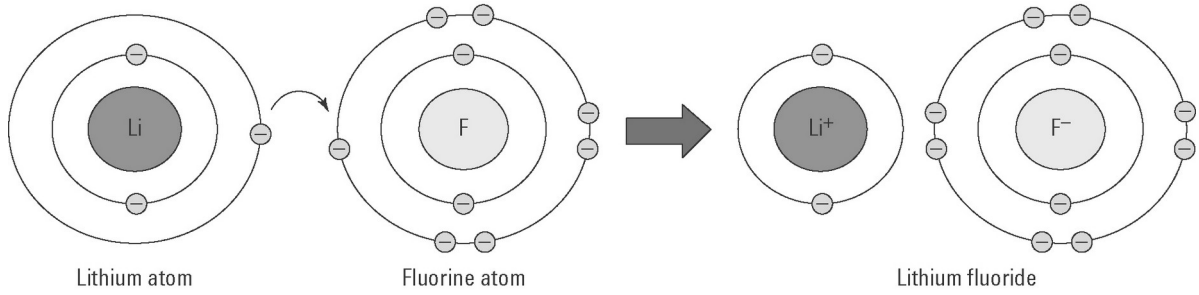
1. When atoms form ions by losing or gaining electrons, they do so to produce a full outer shell, which gives the atom a stable electron configuration. The noble gases are non-reactive because they have a full outer shell.

Complete the following table by showing that each ion has a stable noble gas configuration in its valence (outer) shell.

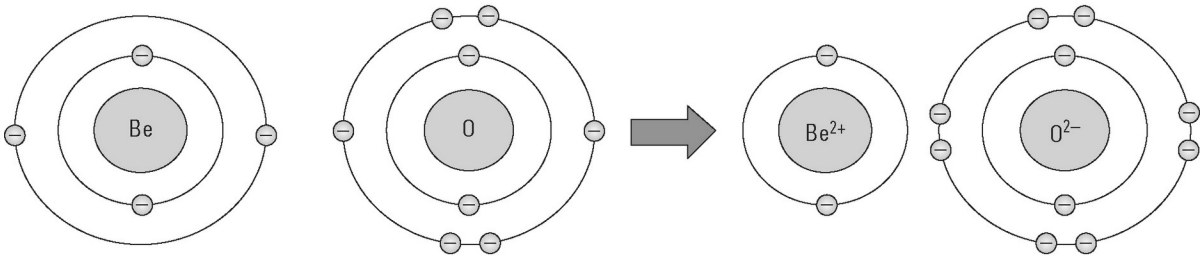
Atom	Electron configuration	Ion	Electron configuration
F	2, 7	F^-	2, 8
N	2, 5	N^{3-}	2, 8
Al	2, 8, 3	Al^{3+}	2, 8
K	2, 8, 8, 1	K^+	2, 8, 8
S	2, 8, 6	S^{2-}	2, 8, 8
Br	2, 8, 18, 7	Br^-	2, 8, 18, 8

Forming an ionic compound

2. The electron shell diagram below shows how a lithium atom and a fluorine atom combine to form lithium fluoride.



Using similar diagrams, show how the compound beryllium oxide forms when a beryllium atom (2, 2) and an oxygen atom (2, 6) react.



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